

BHARGAV REDDY YELICHERLA

Software Developer

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SUMMARY

- Software Developer with 3 years of experience in full-stack web development, backend technologies, IT support, and data science, proficient in **Java**, **Python**, **React.js**, **Spring Boot**, and **AWS**.
- Experienced in designing scalable web applications and **RESTful APIs** using **Flask**, **Node.js**, and **Express.js**, ensuring high performance and secure system communication.
- Expertise in database management with **MongoDB**, **MySQL**, **PostgreSQL**, and **DynamoDB**, optimizing queries for high-traffic applications.
- Skilled in deploying applications on **AWS (EC2, S3, RDS)**, utilizing cloud technologies for scalable solutions and cost-efficient resource management.
- Proficient in machine learning with **TensorFlow**, **Keras**, and **Scikit-learn**, integrating models for data-driven decision-making.
- Built real-time dashboards using **React.js** and **Power BI**, streamlining reporting and enhancing decision-making.
- Strong in **CI/CD** using **Jenkins**, **GitHub Actions**, **Docker**, and **Kubernetes**, automating deployment processes for faster releases.
- Experienced in data engineering, optimizing **ETL** pipelines with **Apache Spark**, **PySpark**, and **Azure Data Factory**.
- Proficient in **Agile** methodologies and **TDD** with **Pytest** and **Jest** for high code quality.

PROFESSIONAL EXPERIENCE

Arizona State University

Software Developer and IT Support

Tempe, AZ, USA

Dec 2023 – Dec 2024

- Led the development of a Comprehensive IT Support and System Optimization Project, designing and deploying **Flask-based RESTful APIs** to automate support workflows, reduce system errors, and improve operational efficiency.
- Implemented robust database schemas with **MongoDB** and **MySQL** for efficient data storage and retrieval, ensuring smooth communication between backend and frontend components.
- Built a **React-based** dashboard integrated with **backend APIs** to track core business KPIs, enhancing real-time decision-making and replacing manual reporting processes.
- Developed and deployed RESTful APIs using **Node.js** and **Express.js**, optimizing backend performance and integrating with **Python-based** services for seamless data flow across systems.
- Automated IT support tasks with **Python scripts**, integrating them into the front-end dashboard for seamless user interactions and reducing repetitive workload by 15%.
- Streamlined integration of unstructured data from 20+ sources into centralized systems, providing backend support for comprehensive analytics and enabling **frontend** visualization of insights.
- Streamlined deployment workflows by implementing **CI/CD** pipelines with **Jenkins** and **GitHub Actions**, automating build, test, and deployment processes, reducing release times and minimizing errors.
- Containerized applications using **Docker**, ensuring consistent environments across development, staging, and production, and orchestrated scalable deployments with **Kubernetes** for better resource management and high availability.
- Developed and deployed machine learning models using **Python** libraries such as **TensorFlow**, **Keras**, and **Scikit-learn**, integrating them into backend services to automate data-driven decision-making processes and improve operational efficiency.
- Deployed backend services on **AWS**, leveraging **EC2** for compute, **S3** for storage, and **RDS** for database management, ensuring high availability, performance, and scalability.
- Applied Test-Driven Development (**TDD**) practices using **Pytest** for backend services and **Jest** for frontend testing, ensuring robust application functionality and reducing defects by 25%.

KPMG

Software Developer

India

Nov 2021 – Dec 2022

- Developed and deployed **RESTful APIs** using **Python** and **Flask**, enabling seamless communication between frontend components and backend databases, improving data exchange efficiency.
- Led the transition to a paperless model by implementing a **Flask-based** electronic booking system, reducing manual effort by 20% and lowering office overhead by 8%.
- Designed and maintained dynamic and responsive front-end components using **HTML**, **CSS**, **JavaScript**, **AngularJS**, **Typescript**, and **Bootstrap**, enhancing user interface efficiency by 15%.
- Automated recurring reporting tasks by developing Flask-based tools, reducing manual reporting efforts by 25% and increasing accuracy.
- Deployed backend services on **AWS**, leveraging **EC2**, **S3**, and **RDS** to enhance application performance, scalability, and ensure high availability for a seamless user experience.

- Optimized backend database queries with **MongoDB** and **PostgreSQL**, reducing system response time by 18%, leading to an improved user experience across platforms.
- Streamlined deployment pipelines using **CI/CD** tools like **Jenkins** and **GitHub Actions**, automating build, test, and deployment workflows, reducing deployment errors and accelerating release cycles.
- Containerized applications using **Docker** and orchestrated scalable deployments with **Kubernetes**, improving resource utilization and system reliability in production environments.
- Enhanced data processing workflows by integrating **Apache Spark** for distributed data processing, achieving a 30% improvement in processing large datasets.
- Applied **TDD** (Test-Driven Development) practices using tools like **Pytest** and **Selenium** to ensure high code quality, reduce bugs by 20%, and improve test coverage.
- Leveraged **Python** libraries like **Matplotlib**, **Plotly**, and **Power BI** to integrate data visualizations within backend systems, enhancing the ability to analyze and present product trends for better decision-making.

Adani

Software Developer Intern

India

March 2021 – Oct 2021

- Coordinated in designing and enhancing front-end interfaces using **HTML**, **CSS**, and **JavaScript**, improving user experience and understanding development best practices.
- Collaborated with the development team to automate small-scale data processing tasks using **Python**, reducing manual efforts by 20%.
- Gained hands-on experience in building and maintaining features for a **MEAN stack** application, contributing to the transition towards a paperless booking system.
- Proficient in **Azure** services, including **Azure Data Factory**, **Azure Storage**, and **Azure SQL**, for cloud-based data engineering and scalable application deployments.
- Engaged in streamlining workflows for electronic business systems, contributing to operational efficiency and cost-reduction efforts.
- Developed interactive dashboards using **Django** frameworks, enabling real-time visualization of key metrics for internal reporting and decision-making.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, JavaScript

Frameworks & Libraries: Flask, Express, Spring Boot, React.js, Angular.js, Django, TensorFlow, Keras, Scikit-learn

Databases: MongoDB, MySQL, DynamoDB, PostgreSQL

Cloud & Deployment: AWS (EC2, S3, Lambda, API Gateway, DynamoDB, IAM), Azure, Docker, Kubernetes

Big Data & ETL: Apache Spark, PySpark, Kafka, Data pipelines, DBT, Azure Data Factory

Data Visualization & BI: Power BI, Tableau

Version Control & CI/CD: Git, GitHub, GitLab, Jenkins, GitHub Actions

Testing & TDD: Pytest, Jest, Selenium

Web Technologies: HTML5, CSS3, JavaScript, TypeScript, Bootstrap

Security & Authentication: JWT, OAuth2

Project Management & Methodologies: Agile (Scrum, Kanban), JIRA

Tools & Platforms: VS Code, PyCharm, JIRA, SAS Studio, RapidMiner

EDUCATION

ARIZONA STATE UNIVERSITY

M.S in Information Technology

Tempe, AZ, USA

Jan 2023 - Dec 2024

PONDICHERY UNIVERSITY

Bachelor of Technology in Computer Science

India

Aug 2018 – May 2022

PROJECTS

Job Application Tracker

developed a full-stack app with Flask and SQL to track job applications, including features for adding, editing, and searching by company, position, and status. Integrated search and filter options, and transitioned to SQL on AWS RDS for efficient database handling. Planned interactive charts for visual progress tracking.

E-Commerce Marketplace

built an e-commerce platform using Node.js, Express.js, and MongoDB with JWT authentication, a dynamic catalog, and shopping cart. Designed a React.js interface for an optimized user experience. Demonstrated full-stack expertise with HTML, CSS, JavaScript, and security features.

Personal Finance Tracker

created a finance tracker to monitor income and expenses, using Tableau for dynamic visualizations and integrating goal-setting for financial milestones, demonstrating full-stack development and user-centric design.